

Week 7 Updates

New architecture I

x = Cartesian particle 4-momenta, ϕ = learnable functions,

$\psi(\cdot) = \text{sgn}(\cdot) \log(|\cdot| + 1)$, $N = \#$ particles, $\|\cdot\|, \langle\cdot\rangle$ = Minkowski norm and inner product

1. Embed scalar time value to random Fourier features through two fully connected layers with 32 and 64 nodes to produce t_{emb}
2. Pass one-hot encoded jet type j_t and number of constituents N through two fully-connected layers with 32 and 64 nodes to produce global embedding g^0
3. Initialize particle cloud $x_1 \dots x_N$, and zero-pad up to 150 if necessary
4. Stack L Lorentz-group equivariant layers
 - 4.1 Input to the $l + 1$ -th layer is $t_{\text{emb}}, g^0, g^L, x_1 \dots x_N$
 - 4.2 Calculate messages $m_{ij}^l = \phi_e^l(\psi(\|x_i^l - x_j^l\|), \psi(\langle x_i, x_j \rangle), g^0, g^l, t_{\text{emb}})$
 - 4.3 Cache the message aggregation

$$\sum_{i,j \in N} \phi_m^l(m_{ij}^l) m_{ij}$$

New architecture II

4.4 Update the global embedding

$$g^{l+1} = \phi_g^l \left(g^0, g^l, t_{\text{emb}} \frac{\alpha_l}{N^2} \sum_{i,j \in N} \phi_m^l(m_{ij}^l) m_{ij} \right)$$

where ϕ_m^l outputs a scalar and α_l is a scalar

4.5 Apply a linear combination to update the 4-vectors

$$x_i^{l+1} = \phi_w^l \left(t_{\text{emb}}, \frac{\beta^l}{N} \sum_{i,j \in N} \phi_m^l(m_{ij}^l) m_{ij} \right) x_i^l + \gamma^l \sum_{j \neq i} \phi_x^l(g^0, g^l, t_{\text{emb}}, m_{ij}) \frac{x_i^l - x_j^l}{1 + \|x_i^l - x_j^l\|}$$

where ϕ_w^l and ϕ_x^l output scalars

4.6 Final output $v_\theta(t, x^0)_i = x_i^L - x_i^0$

References I
